

FROM DESIGN TO INDUCTION: A NEW PARADIGM FOR RESPONDENT-CENTRIC PSYCHOLOGICAL SCALE GENERATION

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ABSTRACT

Psychological scales are essential for diagnosing mental health conditions. Most scales are designed by experts, which is time-consuming and expensive. In this paper, we propose an adaptive respondent-centric psychological scales generation method (PySIG). Inspired by the expert-grounded psychometric framework underlying scale design, we analyze existing scales to construct an item knowledge base, which serves as a foundation to assist in generating customized scales for different respondent groups. To evaluate the quality of the generated scales, we design systematic metrics from psychological perspectives and conduct extensive experiments. Results demonstrate that PySIG can effectively produce scales tailored to respondent characteristics while maintaining alignment with professional psychological constructs.

Index Terms— Psychological Scales Generation, Respondents-Centric Evaluation, Psychological Variables

1. INTRODUCTION

Early detection is crucial for mitigating mental health issues, which are of growing global concern [1]. Psychological scales are efficient, low-cost tools for preliminary mental health assessment [2]. Respondent groups often present diverse mental health conditions, and scales aligned with their living and work environments yield more effective assessments [3]. However, most psychological scales are not tailored to specific groups, highlighting the need for customized tools to address diverse needs.

Traditional scale design depends on expert involvement, making it time-consuming and costly [4]. With advances in intelligent technologies, Automatic Item Generation (AIG) has emerged as a promising alternative that leverages computer technology to streamline test construction [5, 6]. Recent studies leverage large language models (LLMs) for AIG. Götz et al. used GPT-2 to generate items directly from existing psychological tests [7], while Hommel et al. fine-tuned GPT-2 on the IPIP dataset for improved suitability [8, 9]. Lee et al. further employed GPT-3 with personality test items as contextual prompts, enabling more advanced item generation [10]. However, existing studies mainly focus on personality-based items in restricted settings and have not been extended to broader or more complex areas of psychological testing.

Moreover, most approaches still depend on manual expert evaluation, which indicates the need for automated methods to ensure

both efficiency and validity. Recent studies have explored the use of LLMs as substitutes for human evaluation: GPT-3.5 shows high agreement with human judgments [14], and LLMs have been applied to replicate psychological experiments or simulate participants with specific traits [15, 16, 17]. However, these methods are typically task-specific and address only narrow dimensions, making them insufficient for the multi-dimensional and comprehensive evaluation required for psychological scales.

Key challenges in psychological scale generation include verifying the construct validity of generated items and ensuring their suitability for the target respondent groups. These overarching issues can be further specified into three concrete challenges:

1) How to generate personalized scales for different respondent groups? Scale items should be tailored to the respondent group's characteristics, and diversity in items can prevent memorized responses while revealing true conditions [11, 12].

2) How to ensure that the generated scale alignment with psychological principles? Expert-designed scales rely on basic *factors* to ensure items effectively assess specific mental health issues [13], and generated items should satisfy the same criteria.

3) How to conduct intrinsic evaluation of generated psychological scales? While automated methods can efficiently produce large volumes of items, traditional evaluations—such as participant testing or expert review—do not scale to such scenarios. Thus, a systematic and automated evaluation framework is needed to ensure quality.

To address these challenges, we first construct an item knowledge base by inducing psychological variables from expert-designed factors and existing scales, thereby ensuring the professional relevance of the generated items. Second, we propose an adaptive method (PySIG) for generating psychological scale items tailored to specific measurement topics, respondent characteristics, and contextual scenarios. Finally, we design a multi-dimensional evaluation framework to comprehensively assess the quality of the generated scales.

2. METHOD

2.1. Task Definition and Architecture

Given an original scale S , the *primary variable* of the scale ($\mathbf{v}_p$), the *group attributes* C_g , and the *format attributes* C_f that specify the linguistic and structural form of items, the function $f(S, C_g, C_f, \mathbf{v}_p)$ generates a new scale S' that satisfies the structure of S and the conditions C_g and C_f , where I and I' denote the items in S and S' ,

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This work was supported by the National Natural Science Foundation of China (62376138) and the Innovative Development Joint Fund Key Projects of Shandong NSF (ZR2022LZH007).

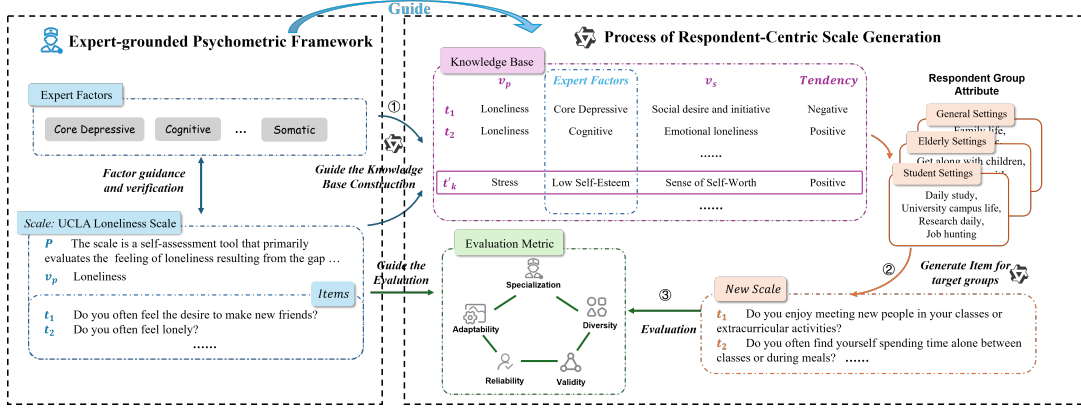


Fig. 1. Architecture of PySIG. The framework consists of three components: (1) construction of the item knowledge base (purple), (2) integration of respondent group attributes (orange), and (3) evaluation and selection of generated scales through designed metrics (green).

respectively.

$$S' = f(S, C_g, C_f, \mathbf{v}_p) = \bigcup_{I_i \in S} P(I_i | \mathbf{v}_p, \mathbf{v}_s, C_g, C_f) P(\mathbf{v}_s | I_i) \quad (1)$$

The generated scales are systematically evaluated by $\text{Eval}()$, and only those satisfying the validity criteria are selected.

$$S^* = \arg \max_{S'} \text{Eval}(S') \quad (2)$$

As shown in Figure 1, our framework consists of three parts. First, we extract sub-variables from items and align them with expert factors to build an item knowledge base, ensuring adherence to the factor structure and professional validity. Second, new items are generated from the knowledge base and group attributes to satisfy specified variables and conditions. Finally, the generated scales are evaluated to ensure quality.

2.2. Psychometrics Guided Item Knowledge Base Construction

In psychology, scales are designed to measure specific psychological traits or states [18], which we define as the *primary variable* ($\mathbf{v}_p$). Experts use factor analysis to uncover latent *expert factors* underlying a scale by analyzing item correlations from large-scale response data [13]. Inspired by this, we refine expert factors into fine-grained *sub-variables* ($\mathbf{v}_s$).

We use existing scales to capture underlying structure and extract control sub-variables. Each item maps to a specific sub-variable, which guides new item generation. we leverage the LLM to infer item-sub-variable mappings.

In psychological scales, respondents answers are converted into numerical scores, with the total score reflecting the level of the target construct. Each item has a scoring *tendency* (T) that can be *positive*, *negative*, or *neutral*. However, respondents may intentionally distort their answers to appear more favorable—for instance, students overstating their “confidence level” in school. When consecutive items share the same positive tendency, such patterns become predictable and easier to manipulate. Introducing a mix of positive and negative items reduces this bias and yields more accurate assessments of true psychological states. We determine the scoring tendency of each item in existing scales and use it as a prompting demonstration to control the tendencies of generated items.

Finally, each item I is stored in the knowledge base together with its primary variable $\mathbf{v}_p$, sub-variable $\mathbf{v}_s$, and scoring tendency T , forming tuples $(I, \mathbf{v}_p, \mathbf{v}_s, T)$.

2.3. Respondent-Centric Psychological Scales Generation

To adapt scales to target populations, items should reflect their living and work environments while preserving the underlying structure. We therefore introduce *group attributes* (C_g) to describe these settings. To enhance diversity, we further define *format attributes* (C_f) to specify statement style and the ratio of scoring tendencies, preventing uniform positive or negative responses.

Based on the *primary variable* ($\mathbf{v}_p$), e.g., *stress*, relevant items are retrieved from the knowledge base, each associated with a sub-variable $\mathbf{v}_s$. The *group attributes* (C_g) describe target subjects and their scenarios, often related to living and work environments—for instance, for *elderly*, scenarios may include *interactions with off-spring, neighbors, or living alone*. Users can specify a batch of scenarios to generate diverse items. The *format attributes* (C_f) define the statement style (e.g., *first-person*) and the *tendency ratio*, indicating the proportion of positively scored items. Overall, $\mathbf{v}_p$, $\mathbf{v}_s$, C_g , and C_f form the LLM prompt to guide the new scales generation.

2.4. Multidimensional Evaluation for Psychological Scales

The evaluation focuses on two aspects: (1) whether the generated items satisfy psychology experts’ requirements, and (2) whether the generated scales are suitable for the target respondents. We propose a set of metrics—*Psychological Specialization*, *Adaptability*, *Diversity*, *Reliability*, and *Validity*—to evaluate the generated scales.

Psychological Specialization. This dimension includes (1) *Primary Variable Specialization* (S_p), evaluating whether items focus on the primary variable, and (2) *Sub-Variable Specialization* (S_ψ), assessing coverage of its sub-variables.

Let $\mathbf{v}_p$ denote the primary variable of scale S . For each generated item $I_i \in S'$, we employ an LLM to check whether I_i is aligned with the variable $\mathbf{v}_p$. The proportion of valid items in S' is then calculated as follow, where $\mathbb{I}(\cdot)$ is the indicator function that equals 1 if the condition holds and 0 otherwise.

$$S_p = \frac{1}{|S'|} \sum_{I_i \in S'} \mathbb{I}(\text{LLM}(I_i, \mathbf{v}_p) = \text{True}) \quad (3)$$

For each generated item $I_i \in S'$, we employ the LLM to predict its sub-variable $Sub(I_i)$. Let x_i denote the original item corresponding to I_i , and $Factor(x_i)$ be its expert-annotated factor. We then construct a co-occurrence matrix $\mathbf{M}_S$ based on the original scale S , where each entry indicates whether a sub-variable and an expert factor co-occur in S . The sub-variable specialization score is then defined as:

$$S_\Psi = \frac{1}{|S'|} \sum_{I_i \in S'} \mathbb{I}(\mathbf{M}_S[Sub(I_i), Factor(x_i)] > 0) \quad (4)$$

Adaptability. Let $\mathcal{A}_j$ be an attribute from $C_g \cup C_f$, which include factors such as scoring tendency, target population, scenario setting, and statement format. For each generated item $I_i \in S'$, we employ the LLM to check whether I_i conforms to the specified attribute $\mathcal{A}_j$. The adaptability score for $\mathcal{A}_j$ is defined as:

$$S_A = \frac{1}{|S'|} \sum_{I_i \in S'} \mathbb{I}(LLM(I_i, \mathcal{A}_j) = True) \quad (5)$$

Diversity. We measure the diversity among items within S' using the Self-BLEU metric $BLEU_{self}$ [19]. A higher Div_{in} score reflects greater diversity.

$$Div_{in} = 1 - \frac{1}{|S'| \times (|S'| - 1)} \times \sum_{I_i \in S'} \sum_{\substack{I_j \in S' \\ I_j \neq I_i}} BLEU_{self}(I_i, I_j) \quad (6)$$

To assess the difference between generated items and the original scale, we compute the BLEU score $BLEU$ [20] for each pair of $I_i \in S'$ and $x_j \in S$. A higher Div_{ex} indicates greater diversity in rewriting compared to the reference items. The score is defined as:

$$Div_{ex} = 1 - \frac{1}{|S'|} \sum_{I_i \in S'} \sum_{x_j \in S} BLEU(I_i, x_j) \quad (7)$$

Reliability. We use prompt engineering to define the roles and psychological states of LLMs, simulating respondents with diverse characteristics to complete the scale. Reliability evaluates the measurement precision, stability, and internal consistency of the scale, which we quantify using Cronbach's α coefficient [21].

Validity. Validity measures the extent to which items reflect the underlying latent variables. In this study, we assess construct validity using the Kaiser-Meyer-Olkin (KMO) measure [22]. Given the one-to-one correspondence between generated and original items, we use LLMs to answer both sets and compute the Pearson correlation [23] to evaluate structural consistency.

Table 1. Target Respondent Groups and Scenario Settings

Group	Scenario Settings
Students	Daily study, university campus life, research activities, job hunting, general daily events
Elderly	Interactions with children, relationships with neighbors, living alone
General	Family life, social life, professional work, academic pursuits

3. EXPERIMENTS AND RESULTS

Datasets. The dataset comprises 19 psychological test scales¹, including 6 mental health scales, 2 behavioral scales, and 11 cognitive

¹<https://github.com/hyc619/PySIG>

attitude scales. Each scale contains the scale title, testing purpose, and specific test items. In the following sections, we will use abbreviations to refer to the titles of the scales.

Experimental Setting. We used Qwen-Max [24] as the base model for item generation and Qwen-Turbo [24] for evaluation to reduce deployment costs. To better simulate human responses, we employed GPT-3.5-turbo as virtual participants.

We select three sets of group attributes C_g (Table 1), use first-person statements for format attributes, and set the negative tendency ratio in C_f to 0.4. For comparison, we adopt a general prompt containing only the reference item and primary variable, differing from our method by excluding sub-variables and specified attributes. Specifically, the baseline prompt instructs the model with the role "You are a psychologist" and asks it to generate a new sentence testing the given primary variable based on a provided reference item, with the output returned in JSON format under the key "new sentence".

Specialization in Generated Scales. We evaluate the professionalism of generated scales using S_P and S_Ψ (Section 2.4), which assess alignment with intended psychological variables. Experiments were conducted on eight psychological scales with available expert factors, comparing PySIG with a baseline using a general prompt template. Results are shown in Figure 2.

The scales generated by PySIG demonstrate strong psychological specialization, with most S_P and S_Ψ scores ranging between 0.8 and 1.0. However, relatively lower S_P scores in SDS [25] and PHQ-9 [26] can be explained by the complexity of depression as the primary variable $\mathbf{v}_p$ —a multifaceted construct where items often capture only partial symptoms. For instance, decreased appetite is one symptom of depression but does not independently indicate the disorder. Although some baseline items align with the primary variable, the lack of domain knowledge leads to insufficient structural validity, reflected in lower S_Ψ scores.

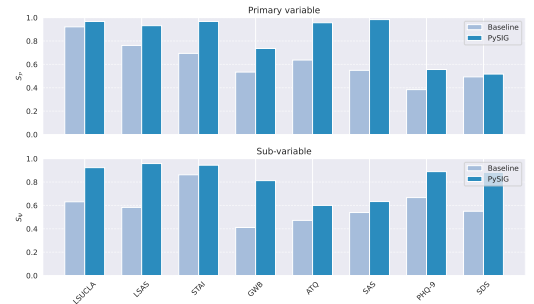


Fig. 2. Psychological Specialization of Generated Scales

Adaptability in Generated Scales. We employ S_A (Section 2.4) to evaluate whether the generated items match group-specific attributes and contextual settings (S_{Target} , S_{Scene}), and whether they adhere to the required formatting and response styles (S_{Tend} , $S_{Statement}$).

As shown in Table 2, the average Adaptability scores across all four aspects exceed 0.88, indicating that the generated items not only align with the contextual needs of the target groups but also exhibit diverse scoring tendencies and statement formats.

Compared to the baseline and original scales, our method produces a more balanced scoring tendency. Specifically, the proportion of negatively scored items is only 5.63% in the baseline and 28.48% in the original scales, whereas PySIG achieves 45.2%. This indicates that our method generates items with more diverse scoring tenden-

Table 2. Adaptability Scores of Generated Psychological Scales

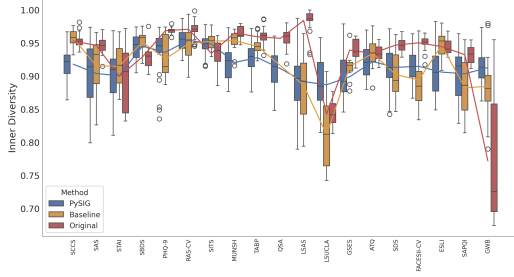
Scale	S_{Tend}		S_{Target}		S_{Scene}		$S_{Statement}$	
	PySIG	Base	PySIG	Base	PySIG	Base	PySIG	Base
GSES	1.000	0.962	0.967	0.527	0.967	0.616	1.000	0.966
SCCS	0.971	0.943	0.905	0.431	0.962	0.523	0.971	0.954
ESLI	0.978	0.972	1.000	0.667	0.944	0.537	1.000	0.974
FACESII-CV	1.000	0.962	0.966	0.386	0.862	0.764	1.000	0.968
SITS	0.906	0.867	0.938	0.467	0.865	0.734	0.927	0.913
ATQ	0.933	0.915	0.978	0.348	1.000	0.837	0.989	0.977
LSUCLA	0.950	0.919	0.967	0.539	0.800	0.638	0.967	0.938
SAS	0.900	0.836	1.000	0.671	0.833	0.817	0.967	0.965
GWB	0.972	0.752	1.000	0.687	1.000	0.674	0.917	0.907
STAI	0.900	0.877	1.000	0.643	0.892	0.650	0.983	0.955
MUNSH	1.000	0.910	0.877	0.533	1.000	0.837	0.860	0.836
LSAS	0.917	0.860	1.000	0.674	0.833	0.649	0.944	0.928
SDS	0.950	0.938	1.000	0.397	0.933	0.672	1.000	0.961
QSA	0.943	0.927	0.782	0.621	0.747	0.599	0.966	0.957
SBDS	0.889	0.867	0.917	0.496	0.917	0.763	0.889	0.834
TABP	0.839	0.764	0.983	0.364	0.883	0.574	1.000	0.941
PHQ-9	0.926	0.894	0.963	0.650	0.741	0.633	0.963	0.935
SAFQI	0.800	0.813	0.850	0.672	0.775	0.496	0.967	0.922
RAS-CV	0.708	0.667	0.875	0.593	0.875	0.652	0.990	0.937
Average	0.920	0.876	0.946	0.545	0.886	0.667	0.963	0.935

cies, which can help prevent respondents from manipulating their answers.

Diversity in Generated Scales. To assess the diversity of generated items both within the scale and relative to the original, we use Div_{in} and Div_{ex} .

For inter-scale discrepancy, PySIG achieves the highest score, indicating that despite incorporating complex user-defined specifications, our method still preserves strong diversity.

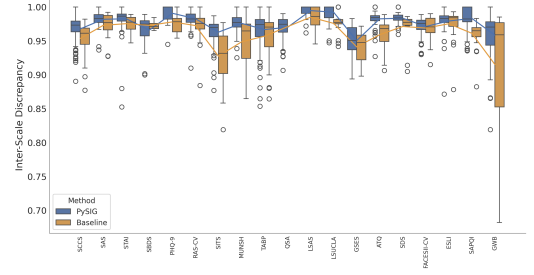
RAS-CV, SITS, and SBDS demonstrate higher inner diversity(Fig.3), while LSAS, LSUCLA, and PHQ-9 exhibit higher inter-scale discrepancy(Fig. 4). Compared to the baseline, PySIG shows slightly lower inner diversity, largely because items often include recurring background or character information introduced by the scene settings. However, the substantive item descriptions remain distinct.

**Fig. 3.** Box plot of Inner Diversity across different scales.

Reliability and Validity in Generated Scales. To ensure statistical validity and reliability, we simulate participant responses using an LLM, with the number of simulated participants set to ten times the number of items in each scale.

Table 3 shows that most generated scales have Cronbach's $\alpha > 0.6$, KMO > 0.7 , significant Bartlett's tests ($p < 0.05$), and Pearson correlations with the original scales > 0.6 , indicating good reliability and structural validity.

Human Evaluation We combined the original scale with a newly generated campus-adapted version and randomly shuffled the items to form a mixed scale. A total of 50 undergraduate and graduate

**Fig. 4.** Box plot of Inter-scale Discrepancy across different scales.**Table 3.** Reliability and validity scores of scales. Cronbach's $\alpha \geq 0.7$, Pearson ≥ 0.6 , and KMO ≥ 0.8 are considered acceptable.

Scale	Reliability	Validity	
	Cronbach's α	Pearson	KMO
FACESII-CV	0.6393	0.7466	0.920
MUNSH	0.4872	0.7155	0.941
ESLI	0.6778	0.6285	0.947
SBDS	0.5536	0.7819	0.882
SITS	0.7147	0.6485	0.963
GSES	0.6278	0.6818	0.790
RAS-CV	0.4649	0.7328	0.900
ATQ	0.6543	0.6113	0.967
SAQ	0.2333	0.7695	0.961
SCCS	0.7114	0.6671	0.780

students (aged 18-27) evaluated Type A behaviors, 50 general adults (aged 28-55) evaluated RAS-CV, and 50 elderly participants (aged over 60) evaluated SAQ. Based on the collected responses, we assessed the reliability and validity of the generated scale and examined its correlation with the original scale.

As shown in Table 4, the generated scale achieved a Cronbach's α exceeding 0.83, indicating high internal consistency. Regarding structural validity, the Kaiser-Meyer-Olkin (KMO) measure was above 0.7, and Bartlett's test of sphericity was statistically significant ($p < 0.05$), confirming the suitability of the data for factor analysis. Moreover, the Pearson correlation between scores of the generated and original scales exceeded 0.75, suggesting strong structural consistency.

Table 4. Human evaluation results across different populations

Population	Reliability	Validity		
	Cronbach's α	KMO	Significance	Pearson
Student	0.841	0.794	0.001	0.752
General	0.831	0.763	0.001	0.786
Elderly	0.847	0.802	0.002	0.769

4. CONCLUSION

In this paper, we propose the psychology control variables induced method on psychological scales generation, which are caught from the existing scales. By enabling the LLM, we construct the psychological knowledge base with which our model can generate multiple psychological items that are specially tailored for any target population and scenario. To assess the quality of items, we design a set of systematic metrics from the point of view of psychology and perform extensive experiments to verify our approach.

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